Sequences — problem 9.25

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The sequence x_n is defined by $x_1 = 1/2$, $x_{k+1} = x_k^2 + x_k$. Find the integer part of the sum

$$\frac{1}{x_1 + 1} + \frac{1}{x_2 + 1} + \cdots + \frac{1}{x_{100} + 1}.$$